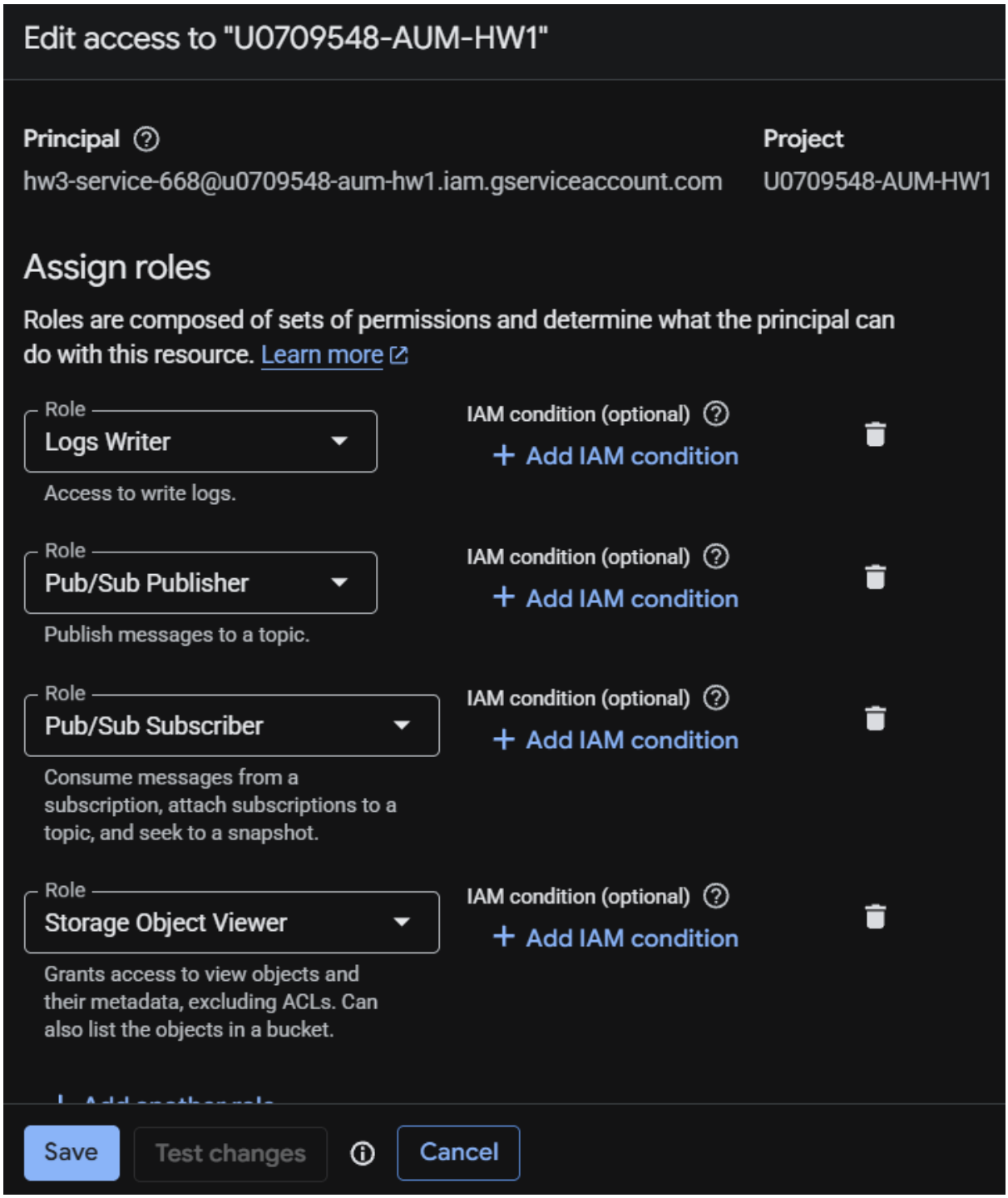
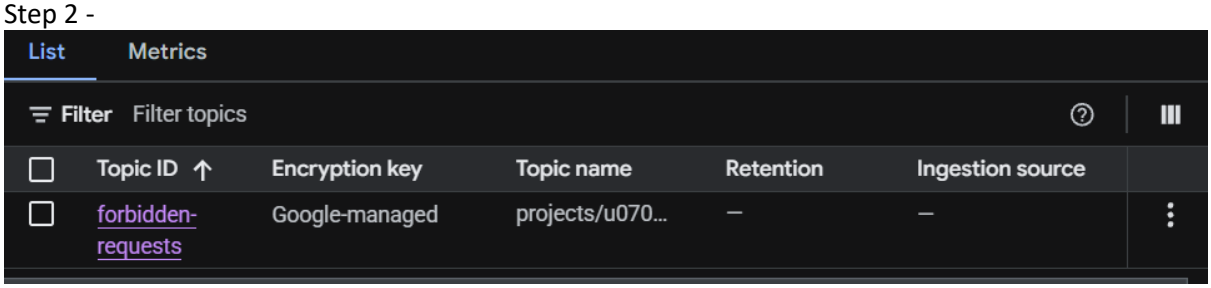
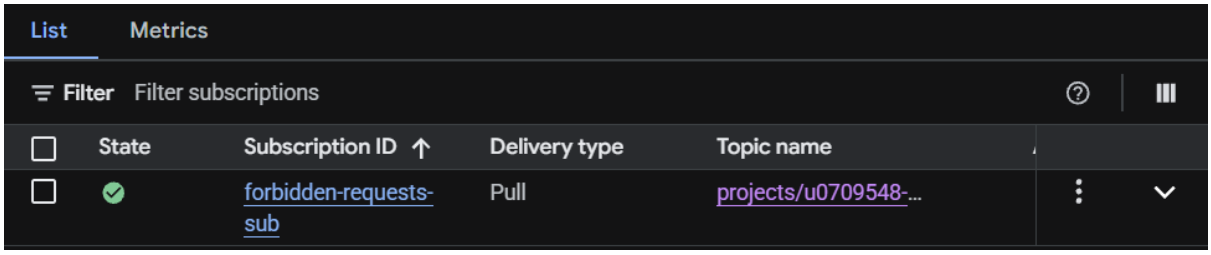
AUM GHELANI _ HW 3 _ CLOUD COMPUTING _ U07079548
(Question order is not steps order - Questions are answered along with the steps in the Screenshots)
Step 1 – Create a Service Account and Grant roles - Storage Object Viewer, Pub/Sub Publisher, Pub/Sub Subscriber, Logging Log Writer

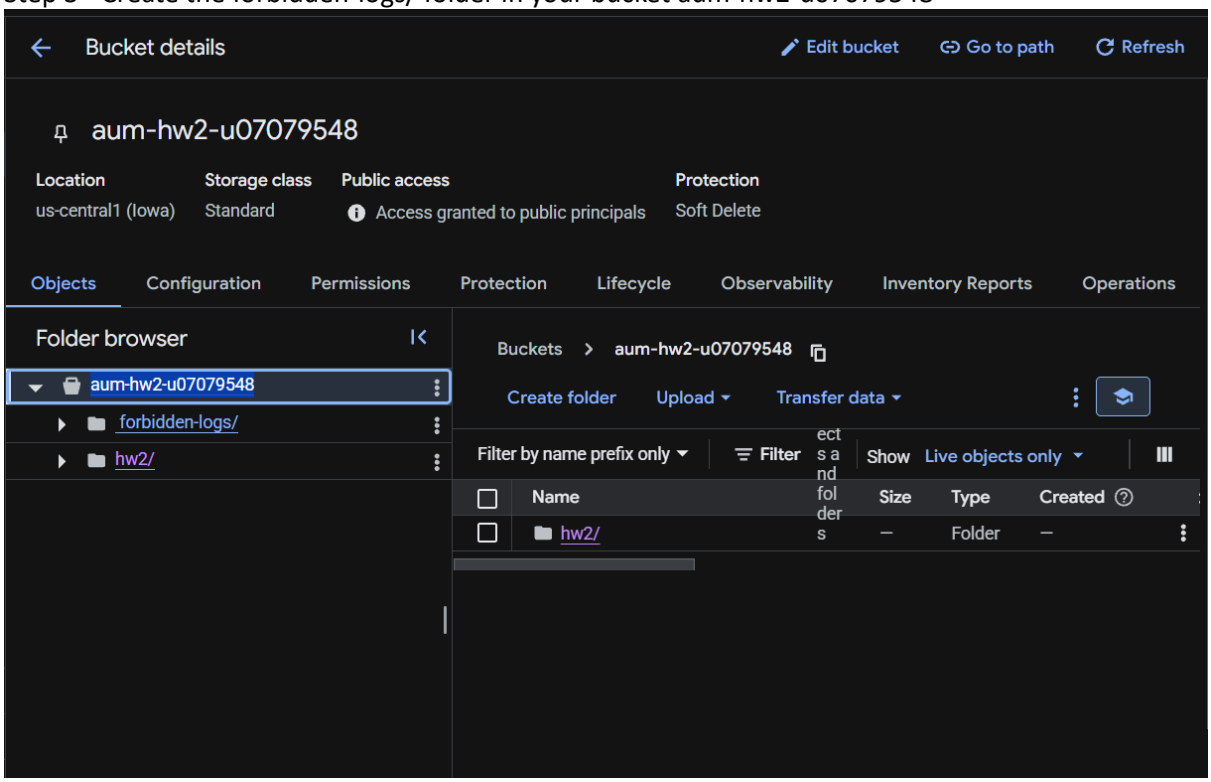


Step 2 -

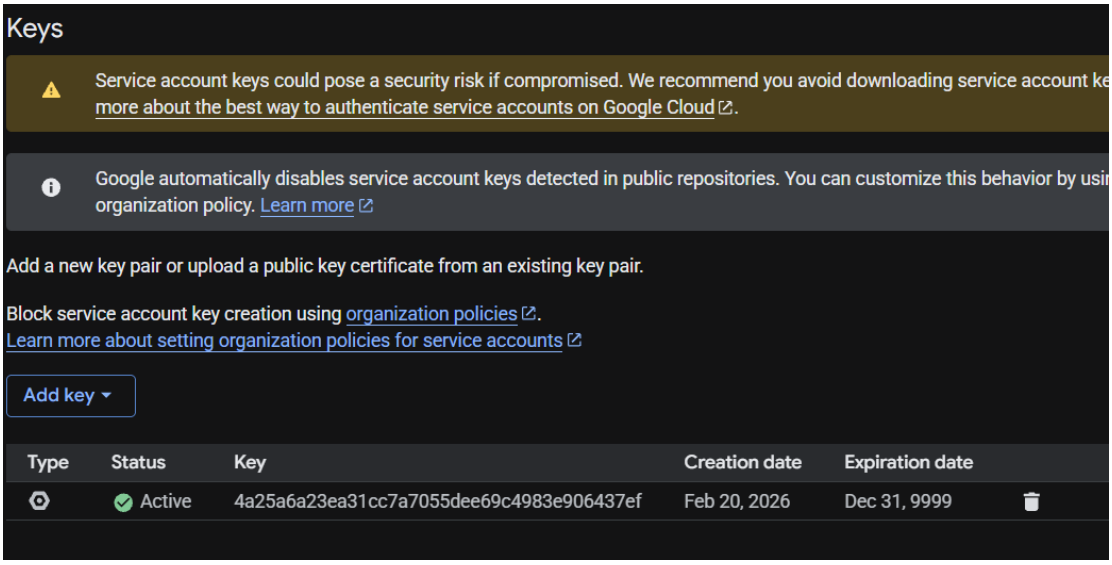




Step 3 - Create the forbidden-logs/ folder in your bucket aum-hw2-u07079548



Step 4 - Download a Service Account Key (Still not at impersonation yet)



```
PROBLEMS    OUTPUT    TERMINAL    DEBUG CONSOLE                                pwsh

PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3> set GOOGLE_APPLIC
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3> set GOOGLE_APPLIC
ATION_CREDENTIALS=D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3\u07
Set-Variable: A positional parameter cannot be found that accepts argume
nt '528'.
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3> $env:GOOGLE_APPLI
CATION_CREDENTIALS="D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3\u
0709548-aum-hw1-4a25a6a23ea3.json"
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3>
```

Step 5 – Changed the pubsub project id in both files

```
# ── Configuration ──────────────────────────────────────────────
BUCKET_NAME        = "aum-hw2-u07079548"
FILE_PREFIX        = "hw2/"
PUBSUB_PROJECT_ID  = "U0709548-AUM-HW1"
PUBSUB_TOPIC_ID    = "forbidden-requests"
```

```
# ── Configuration ──────────────────────────────────────────────
GCP_PROJECT_ID     = "U0709548-AUM-HW1"
SUBSCRIPTION_ID    = "forbidden-requests-sub"
BUCKET_NAME        = "aum-hw2-u07079548"
LOG_PREFIX         = "forbidden-logs/"          # separate directory in the bucket
LOG_BLOB_NAME      = f"{LOG_PREFIX}forbidden_requests.log"
```

Step 6 -

```
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3> gcloud functi
ons deploy hw2 --gen2 --runtime=python313 --region=us-central1 --sou
rce=. --entry-point=handle_request --trigger-http --allow-unauthenti
cated --service-account=hw3-service-668@u0709548-aum-hw1.iam.gservic
eaccount.com --project=u0709548-aum-hw1
Preparing function...done.
X  Updating function (may take a while)...
  OK [Build] Logs are available at [https://console.cloud.google.co
  m/cloud-build/builds;region=us-central1/56ac5164-8fee-439c-b2a2-7
  a5cd99c2fc2?project=1097067927322]
     [Service]
  .  [ArtifactRegistry]
  .  [Healthcheck]
  .  [Triggercheck]
Completed with warnings:
  [INFO] A new revision will be deployed serving with 100% traffic.
You can view your function in the Cloud Console here: https://consol
e.cloud.google.com/functions/details/us-central1/hw2?project=u070954
8-aum-hw1

buildConfig:
  automaticUpdatePolicy: {}
  build: projects/1097067927322/locations/us-central1/builds/56ac516
4-8fee-439c-b2a2-7a5cd99c2fc2
  dockerRegistry: ARTIFACT_REGISTRY
  dockerRepository: projects/u0709548-aum-hw1/locations/us-central1/
repositories/gcf-artifacts
```

```
repositories/gcf-artifacts
  entryPoint: handle_request
  runtime: python313
  serviceAccount: projects/u0709548-aum-hw1/serviceAccounts/10970679
27322-compute@developer.gserviceaccount.com
  source:
    storageSource:
      bucket: gcf-v2-sources-1097067927322-us-central1
      generation: '1771643882914387'
      object: hw2/function-source.zip
  sourceProvenance:
    resolvedStorageSource:
      bucket: gcf-v2-sources-1097067927322-us-central1
      generation: '1771643882914387'
      object: hw2/function-source.zip
createTime: '2026-02-21T02:36:22.732264646Z'
environment: GEN_2
labels:
  deployment-tool: cli-gcloud
name: projects/u0709548-aum-hw1/locations/us-central1/functions/hw2
satisfiesPzi: true
serviceConfig:
  allTrafficOnLatestRevision: true
  availableCpu: '0.1666'
  availableMemory: 256M
```

```
  availableMemory: 256M
  environmentVariables:
    LOG_EXECUTION_ID: 'true'
  ingressSettings: ALLOW_ALL
  maxInstanceCount: 12
  maxInstanceRequestConcurrency: 1
  revision: hw2-00004-daj
  service: projects/u0709548-aum-hw1/locations/us-central1/services/
hw2
  serviceAccountEmail: hw3-service-668@u0709548-aum-hw1.iam.gservice
account.com
  timeoutSeconds: 60
  uri: https://hw2-zgn3xxujtq-uc.a.run.app
state: ACTIVE
updateTime: '2026-02-21T03:18:50.163739002Z'
url: https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3>
```

Step 7 – Running Pub Sub Service

```
PS D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3> $env:GOOGLE_A
PPLICATION_CREDENTIALS = "D:\Masters Studies\CS 528 - Cloud Computin
g\HW\HW3\u0709548-aum-hw1-4a25a6a23ea3.json"
>> python "D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3\servic
e2.py"
>>
D:\Masters Studies\CS 528 - Cloud Computing\HW\HW3\service2.py:1: Sy
ntaxWarning: invalid escape sequence '\s'
  """
[Service 2] Listening on projects/U0709548-AUM-HW1/subscriptions/for
bidden-requests-sub …
```

Step 8 –

```
PS C:\WINDOWS\system32> curl.exe -i "https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw3-service1?file=hello.txt"
HTTP/1.1 200 OK
content-type: text/html; charset=utf-8
x-cloud-trace-context: c0ac93011ac470109b88ad5873d9a37c;o=1
date: Sat, 21 Feb 2026 01:37:07 GMT
server: Google Frontend
Content-Length: 0
Alt-Svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

PS C:\WINDOWS\system32> curl.exe -i -H "X-country: Iran" "https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw3-service1?file=hello.txt"
HTTP/1.1 400 Bad Request
content-type: text/html; charset=utf-8
x-cloud-trace-context: ce22546f4124e117111e4f3bf9dd7d42
date: Sat, 21 Feb 2026 01:37:17 GMT
server: Google Frontend
Content-Length: 45
Alt-Svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

Permission Denied: export restrictions apply.
PS C:\WINDOWS\system32> curl.exe -i "https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw3-service1?file=this_is_missing.txt"
HTTP/1.1 404 Not Found
content-type: text/html; charset=utf-8
x-cloud-trace-context: d9aa7623fd6c96bf887d06a979451fd7;o=1
date: Sat, 21 Feb 2026 01:37:27 GMT
server: Google Frontend
Content-Length: 14
Alt-Svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

File Not Found
PS C:\WINDOWS\system32> curl.exe -i -X POST -H "Content-Length: 0" "https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw3-service1?file=hello.txt"
HTTP/1.1 501 Not Implemented
content-type: text/html; charset=utf-8
x-cloud-trace-context: d50d0e037129bcb4a983496282a94928
```

- 100 Request httpclient.exe file

Logs –

88% of results are similar and can be hidden. Hide similar entries Preview

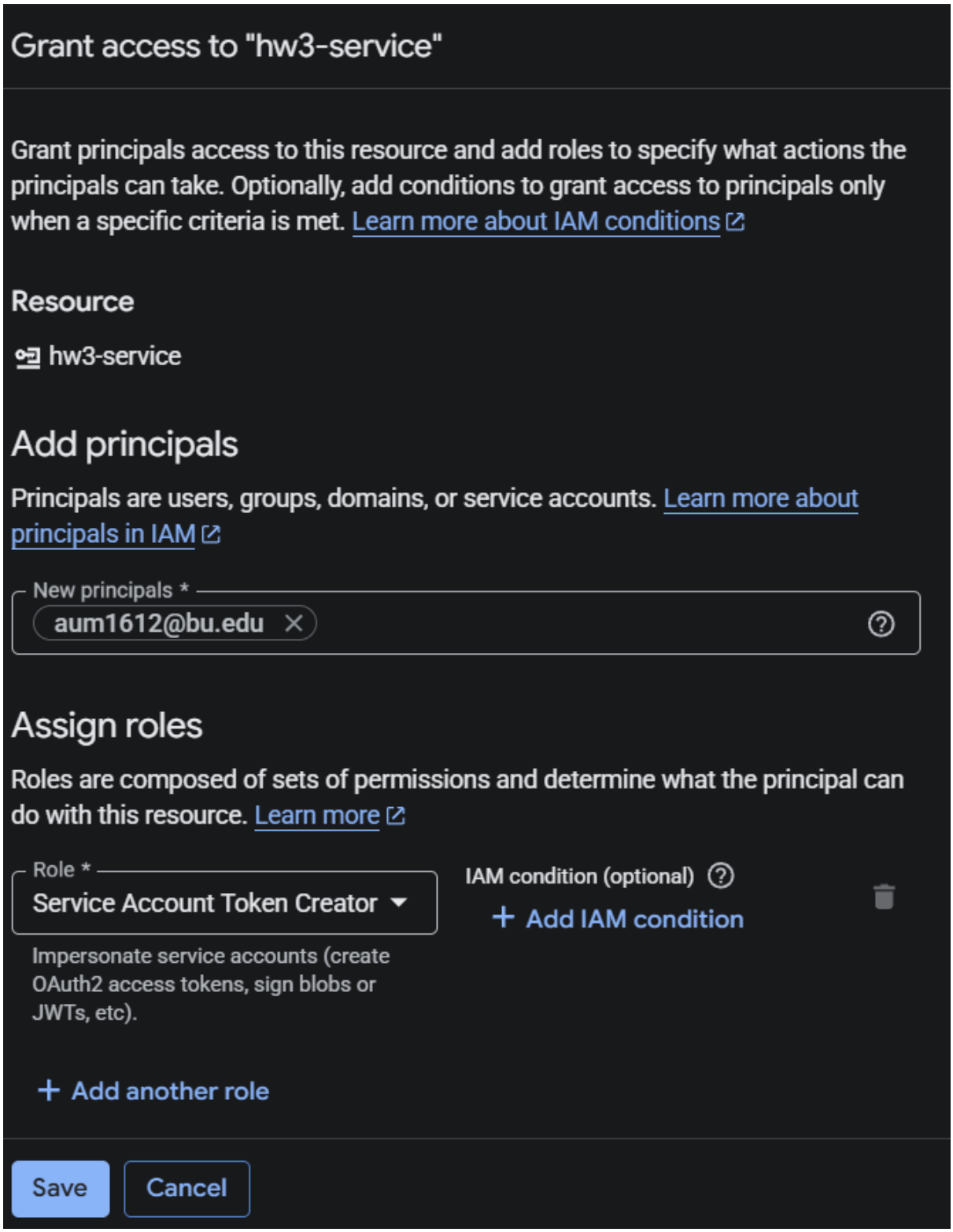
2026-02-20 22:02:36.521	GET	200	20.75 KiB	109 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/7727.html	
2026-02-20 22:02:36.757	GET	200	148.56 KiB	98 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/678.html	
2026-02-20 22:02:37.077	GET	400	174 B	4 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/8594.html	
2026-02-20 22:02:37.081	{"country":"Iraq", "event":"FORBIDDEN_COUNTRY", "file_requested":"8594.html", "remote_addr":"169.254.169.126"}					
2026-02-20 22:02:37.202	GET	200	40.34 KiB	84 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/2788.html	
2026-02-20 22:02:37.447	GET	200	46.7 KiB	98 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/1041.html	
2026-02-20 22:02:37.701	GET	200	61.39 KiB	80 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/1926.html	
2026-02-20 22:02:37.949	GET	200	38.37 KiB	88 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/6141.html	
2026-02-20 22:02:38.210	GET	200	170.1 KiB	77 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/672.html	
2026-02-20 22:02:38.525	GET	200	112.82 KiB	99 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/6449.html	
2026-02-20 22:02:38.824	GET	200	86.84 KiB	94 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/885.html	
2026-02-20 22:02:39.101	GET	200	105.96 KiB	115 ms	https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw2/9006.html	

2026-02-20 22:18:47.890	Cloud Run	ReplaceInternalService	hw2-00004-daj	Ready condition status changed to True for Revision hw2-00004-daj with message: Deploy…
2026-02-20 22:18:49.869	Cloud Run	ReplaceInternalService	hw2	Ready condition status changed to True for Service hw2.
2026-02-20 22:18:50.183	Cloud Functions	UpdateFunction	us-central1:hw2	aum1612@bu.edu audit_log, method: "google.cloud.functions.v2.FunctionService.UpdateF…
2026-02-20 22:18:52.431	Cloud Run	SetIamPolicy	us-central1:hw2	aum1612@bu.edu audit_log, method: "google.cloud.run.v1.Services.SetIamPolicy", principal_ema…

Last Part of 100 Request


```
ptate velit esse\r\ncillum dolore eu fugiat nulla pariatur. Excepteu
r sint occaecat cupidatat non proident, sunt in culpa qui officia de
serunt mollit anim id est laborum.\r\n<p>\r\n<a HREF="13947.html"> T
his is a link </a>\r\n<p>\r\nLorem ipsum dolor sit amet, consectetur
 adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolo
re magna aliqua. Ut enim ad\r\nminim veniam, quis nostrud exercitati
on ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aut
e irure dolor in reprehenderit in voluptate velit esse\r\ncillum dol
ore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non
proident, sunt in culpa qui officia deserunt mollit anim id est labo
rum.\r\n<p>\r\n<a HREF="8003.html"> This is a link </a>\r\n<p>\r\nLo
rem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmo
d tempor incididunt ut labore et dolore magna aliqua. Ut enim ad\r\n
minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliq
uip ex ea commodo consequat. Duis aute irure dolor in reprehenderit
in voluptate velit esse\r\ncillum dolore eu fugiat nulla pariatur. E
xcepteur sint occaecat cupidatat non proident, sunt in culpa qui off
icia deserunt mollit anim id est laborum.\r\n<p>\r\n<a HREF="18312.h
tml"> This is a link </a>\r\n<p>\r\n</body>\r\n</html>\r\n'
```

IMPERSONATION (WITHOUT KEY)



Service Account Impersonation via google.auth.impersonated_credentials
My personal Google account (authenticated via gcloud auth login) acts as the source credential. The Python SDK calls GCP's IAM API to exchange my personal token for a short-lived token scoped to the hw3-service-668 service account. This token expires automatically and is refreshed as needed.
Why I chose it over key files: No private key is stored on disk — eliminating the risk of credential leakage. Why not gcloud auth application-default login: that grants my full user permissions, which are far broader than what the service account needs (only Pub/Sub subscribe + Storage write).

After Impersonation

```
C:\Users\Aum\AppData\Roaming\Python\Python313\site-packages\google\a
uth\_default.py:114: UserWarning: Your application has authenticated
 using end user credentials from Google Cloud SDK without a quota pr
oject. You might receive a "quota exceeded" or "API not enabled" err
or. See the following page for troubleshooting: https://cloud.google
.com/docs/authentication/adc-troubleshooting/user-creds.
  warnings.warn(_CLOUD_SDK_CREDENTIALS_WARNING)
[Service 2] Authenticated via impersonation as: hw3-service-668@u070
9548-aum-hw1.iam.gserviceaccount.com
[Service 2] Listening on projects/U0709548-AUM-HW1/subscriptions/for
bidden-requests-sub …
```

```
PS C:\WINDOWS\system32> curl.exe -i "https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw3-service1?file=hello.txt"
HTTP/1.1 200 OK
content-type: text/html; charset=utf-8
x-cloud-trace-context: f5fe741ad8bf96795ea0286529372f51;o=1
date: Sat, 21 Feb 2026 03:48:22 GMT
server: Google Frontend
Content-Length: 0
Alt-Svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

PS C:\WINDOWS\system32> curl.exe -i -H "X-country: Iran" "https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw3-service1?file=hello.txt"
HTTP/1.1 400 Bad Request
content-type: text/html; charset=utf-8
x-cloud-trace-context: 2ef39fd029d928d86848f77f7e8092e6
date: Sat, 21 Feb 2026 03:48:27 GMT
server: Google Frontend
Content-Length: 45
Alt-Svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

Permission Denied: export restrictions apply.
PS C:\WINDOWS\system32> curl.exe -i "https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw3-service1?file=this_is_missing.txt"
HTTP/1.1 404 Not Found
content-type: text/html; charset=utf-8
x-cloud-trace-context: 7876b675de087f322093b2e6ace59edc;o=1
date: Sat, 21 Feb 2026 03:48:34 GMT
server: Google Frontend
Content-Length: 14
Alt-Svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

File Not Found
PS C:\WINDOWS\system32> curl.exe -i -X POST -H "Content-Length: 0" "https://us-central1-u0709548-aum-hw1.cloudfunctions.net/hw3-service1?file=hello.txt"
HTTP/1.1 501 Not Implemented
content-type: text/html; charset=utf-8
x-cloud-trace-context: ab0554265191744692bdbe0143c3a7d1
date: Sat, 21 Feb 2026 03:48:38 GMT
server: Google Frontend
Content-Length: 22
Alt-Svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

Method Not Implemented
```

Log after impersonation

```
> 2026-02-20 22:18:47.890  Cloud Run  ReplaceInternalService  hw2-00004-daj  Ready condition status changed to True for Revision hw2-00004-daj with message: Deploy…
> 2026-02-20 22:18:49.869  Cloud Run  ReplaceInternalService  hw2  Ready condition status changed to True for Service hw2.
> 2026-02-20 22:18:50.183  Cloud Functions  UpdateFunction  us-central1:hw2  aum1612@bu.edu  audit_log, method: "google.cloud.functions.v2.FunctionService.UpdateF…
> 2026-02-20 22:18:52.431  Cloud Run  SetIamPolicy  us-central1:hw2  aum1612@bu.edu  audit_log, method: "google.cloud.run.v1.Services.SetIamPolicy", principal_ema…
```